TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

# Dismantling and Reassembling of Loom Parts

*The discipline of taking a mechanism apart and returning it to specification: sequence, marking, inspection, clearances, re-timing — practised on the rapier drive and gripper head, the take-up gear train and the shedding cam unit.*

FM-4102 · ADVANCED WEAVING TECHNOLOGY-II SESSIONAL · EXPERIMENT 2 · FABRIC MANUFACTURING SPECIALIZATION

## 1. Aim and theory

**Aim:** to dismantle selected loom sub-assemblies, inspect their parts for wear and damage, and reassemble them to the maker's settings, verifying timing and function afterwards.

Dismantling is a reading exercise: a mechanism states its condition in wear patterns — polished tracks where balls or rollers run, backlash grown between gear teeth, fretting at keys and splines, tired springs, contaminated lubricant. The experiment's rules exist to protect the evidence and the machine: **one assembly at a time; parts laid out in removal order on a clean tray; every timing or adjustment position marked before it is disturbed; nothing forced — a part that will not move is telling you about a fastener or a sequence you have missed.** Reassembly is the reverse discipline plus settings: clearances, end-floats and timings are *restored to numbers*, not to "about where it was".

## 2. Apparatus and tools

- Loom (isolated), the sub-assemblies released for the practical: rapier drive unit and gripper head, take-up gear train, shedding cam/dobby drive element (as assigned).
- Ring and socket spanners, Allen keys, circlip pliers, soft-faced mallet, puller set, feeler gauges, dial indicator with magnetic stand, vernier caliper, torque wrench, cleaning solvent and lint-free cloth, fresh lubricant to the maker's specification, parts tray, marker and tags, camera/phone for sequence photographs.

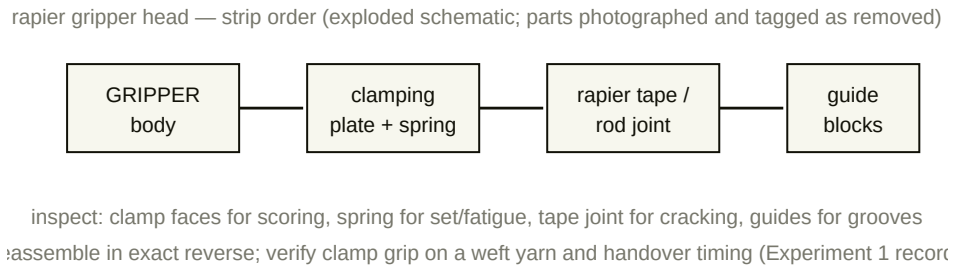



Figure 1 — Strip map of the rapier gripper head used in this experiment: the drawing made *before* dismantling, against which the tray is laid out and the rebuild checked.

## 3. Procedure

1. **Before touching:** photograph the assembly in place; mark timing relations (scribe or paint-mark gear teeth in mesh, shaft-to-coupling positions); record existing settings — backlash readings (dial indicator), end-float (feeler/dial), spring lengths; note the loom's angle on the timing disc if the assembly is timed.
2. **Strip in sequence:** release fasteners evenly (cross-pattern on covers — never one side fully first); withdraw shafts with pullers, not levers against machined faces; lay parts on the tray *in removal order, oriented as fitted*; tag anything whose orientation is not self-evident. Keep each assembly's fasteners with that assembly.
3. **Clean and inspect:** wash parts, dry, and inspect against the checklist: gear teeth (pitting, scoring, tooth thinning), bearings (roughness, play), cams and followers (track polish vs pitting, roller freedom), springs (free length vs specification, set), gripper clamp faces (scoring, spring force by sample weft pull test), tapes/rods (cracks at joints), seals and lubricant state. Record findings per part: serviceable / worn-within-limit / replace.
4. **Reassemble:** exact reverse sequence with fresh lubricant where specified; fasteners to torque in cross-pattern; restore the marked timing relations; set clearances to the maker's figures and record the achieved values (backlash, end-float).
5. **Verify:** turn the machine by hand through several complete cycles watching the rebuilt assembly through its timing events (compare with the Experiment 1 timing record); then run at low speed, then at working speed; weave a length of cloth and inspect it (Experiment 3's eye) for faults traceable to the rebuilt unit — e.g. gripper faults as missing/short picks, take-up faults as pick-spacing bars.

## 4. Worked calculation (model data)

> **BACKLASH CHECK ON THE TAKE-UP GEAR TRAIN**

Method: lock the input shaft; rock the output gear and read total movement at the pitch radius on the dial indicator.
Maker's specification for the pair: backlash 0.10–0.20 mm at the pitch circle. Measured on the stripped train (before cleaning): **0.34 mm** — outside limit; inspection found the smaller gear's teeth thinned by wear (the classic pattern: the pinion, cycling many times per output revolution, wears fastest).
After fitting the replacement pinion and re-meshing: measured **0.15 mm** — inside specification, recorded.
Why it matters in cloth: take-up backlash is lost motion at every direction change of load; it appears on the fabric as irregular pick spacing — a mechanism measurement converted, by one line of reasoning, into a fabric defect (Experiment 3).

## 5. Inspection record (model data — form to be completed per assembly)

| PART | CHECK | FINDING (MODEL) |
|---|---|---|
| Gripper clamp spring | Free length; grip on sample weft | Length at lower limit; grip weak → spring replaced |
| Rapier tape joint | Cracks, rivet tightness | Serviceable; cleaned and re-fitted |
| Take-up pinion | Tooth thickness; backlash pair reading | Worn (0.34 mm pair backlash) → replaced; re-set 0.15 mm |
| Shedding cam follower roller | Freedom, track pitting | Roller free; track polished, no pitting → serviceable |
| Bearings (drive shaft) | Play, roughness by hand | Smooth, no perceptible play → serviceable, re-greased |

## 6. Precautions

- Machine isolated and locked out before any cover is opened; stored energy (springs under compression, raised sley) released or restrained before release of its fasteners.
- No hammering on shafts, races or gear teeth — pullers and soft drifts only; bearing races never driven through the rolling elements.
- Solvent used away from ignition, parts fully dry before lubricant is applied; old and new lubricants never mixed blindly.

- If a reassembly measurement cannot be brought into specification, the machine is **not** run — the discrepancy is reported. An experiment that ends "it would not set" has still succeeded if the record is honest.

## 7. Practice (viva preparation)

1. Why are timing marks made *before* dismantling rather than found again during reassembly?
2. A gear pair's specified backlash is 0.10–0.20 mm. Two rebuilds measure 0.05 mm and 0.30 mm. Reject both — explain each failure mode.
3. State the inspection difference between a polished cam track and a pitted cam track — which is acceptable?
4. Why do pinions in a reduction train usually wear faster than the wheels they drive?
5. Give the cloth-level symptom that would make you suspect (a) gripper clamp wear, (b) take-up backlash, after a rebuild.

### Answer key

1. Because mesh and phase relations are invisible once separated: a gear can re-mesh one tooth out and still turn, but every downstream timing (Experiment 1's angles) shifts by that tooth. Marks preserve the machine's calibrated state as evidence; re-finding timings from scratch is slow and error-prone.
2. **0.05 mm (too tight):** teeth bind as they heat and expand; scuffing, noise, rapid wear and possible seizure. **0.30 mm (too loose):** lost motion — impact loading at each reversal, noisy running, and (in take-up) irregular pick spacing. Backlash is a window, not a minimum.
3. Polish is uniform wear-in of contacting surfaces — normal and acceptable if dimensions hold. Pitting is fatigue failure: small craters where subsurface cracks have broken out — the surface is dying and will shed debris into the lubricant; the part is condemned or life-limited.
4. Each pinion tooth meshes once per pinion revolution, and the pinion makes (ratio) times more revolutions than its wheel: in a 4:1 pair the pinion's teeth each do four times the work cycles of the wheel's, in the same running hours — so it reaches its wear limit first.
5. (a) **Missing or short picks, weft tails at the selvedge** — the clamp is dropping or slipping the weft at handover. (b) **Pick-spacing irregularity/weft bars in bands** — lost motion in the cloth take-up path. Both send you back to the tray, not to the loom settings.

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